

PRANAV DESHPANDE

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SUMMARY

An accomplished and results-driven senior software engineer with extensive experience in AI/ML systems, scalable platform engineering, and cloud-native microservices architecture. Designs and delivers end-to-end generative AI platforms to automate complex enterprise workflows. Delivers measurable business impact, compressing timelines and driving revenue across environments.

EXPERIENCE

WALMART, Sunnyvale, California

Senior Software Engineer, 2022 – Present

Walmart – Agentic AI for fashion

- Led the architecture of Walmart's Agentic AI platform for fashion design. Compressing the design-to-shelf cycle from 53 to 8 weeks (85% reduction) and driving 15K+ units sold in 3 week period (best seller)
- Owned the trend intelligence pipeline ingesting 50K+ fashion signals from social media. Surfacing data backed actionable trends to guide product design decisions and market positioning.
- Architected a multi-model LLM orchestration layer with agentic workflows, microservices on native cloud infra with RBAC authentication. Enabling 15+ private brands to automate product design from trend signal to shelf-ready output without manual handoffs
- Established the AI evaluation framework for contextual drift detection. Improving platform output accuracy from 75% to 93%, reducing incorrect product decisions and strengthening stakeholder trust
- Technologies / Expertise: Python, LangChain, LangGraph, FastAPI, React, TypeScript, Milvus, PostgreSQL, Kubernetes, Azure OpenAI, Google Gemini, Airflow, Brandwatch, Trendalytics, LLM-as-Judge, VLM-as-Judge

Sam's Club — Membership & Payments Platform

- Led zero-downtime migration of PCI-compliant call center payment app from legacy cloud to Walmart native cloud, eliminating outage risk from recurring certificate expirations
- Resolved blockers across 5+ teams, Walmart's first non-PCI-to-PCI communication documentation on legacy systems.
- Drove point-of-sale parity initiative for \$22.5M/year in membership sign-ups (\$750K per club), involving authentication layer, location-based club data aggregation, and membership upgrade/downgrade workflows
- Owned the tax exemption platform for business owners. Automated multi-state compliance workflows, deployed across Sam's Club locations, and delivered 7 days ahead of schedule
- Technologies / Expertise: Node.js, Azure Authentication, Grafana, GSLB, GraphQL, Kubernetes, PingFed

Sam's Club — Scan and Go

- Designed monitoring platform for mobile logs across 30+ Sam's Club locations and 5000+ devices, proactively detect silent failures before they impact customer transactions.
- Eliminated secret exposure, improved developer productivity by migrating 150+ credentials, enabled auto fetching pipelines and strengthening PCI compliance posture across the payment platform
- Real-time telemetry pipeline ingesting 1.5B Kafka messages across 672 partitions at 3.5 GB per hour
- Technologies / Expertise: Kafka, HashiCorp Vault, Akeyless, Kubernetes, External Data Pipelines

WALMART, Sunnyvale, California

Software Engineer 3, 2019 – 2022

Scan and Go — Platform Engineering

- Designed and implemented orchestrator components in massively scalable, multi-tenant platforms on Node.js and Oracle, handling high-throughput transaction processing. 600-1000 OPM. 300k – 900k transactions per day.
- Spearheaded production migration of platform, cache, and Kafka infrastructure from one cloud environment to another, integrating dual-cloud architecture and reducing data leaks and security threats.
- Redesigned automated logging library, achieving 15% log volume compression across platform.
- Led CCPA privacy compliance implementation across 35+ California Sam's Club locations, protecting member records.
- Migrated membership services to cloud, improving platform availability and reducing response time by 25%.
- Implemented machine learning image recognition for Innovation Club (NOW Club), enabling checkout via image recognition across 4 pilot clubs
- Technologies / Expertise: Node.js, Oracle, MeghaCache, Kafka, Machine Learning, Custom Node.js Libraries

EKRYP, Tempe, Arizona

Machine Learning Intern, 2019

- Developed failure prediction model across 3,700 products from 12 customers for proactive maintenance.
- Built AI-NLP based problem identification system for technicians, reducing expert engineer escalations by ~25%.

CYR3CON, Tempe, Arizona

Data Engineering Intern, 2018

- Built end-to-end data leakage detection system ensuring 100% data integrity across pipelines.
- Developed attack pattern prediction system analyzing 31,000+ CVEs to expose potential security threats.
- Spearheaded data mapping across MongoDB and 18 API interfaces, alerting development teams to inconsistencies.
- Targeted 40 data scrapers / crawlers on dark net websites using Python and Selenium.
- Maintained 92% crawler uptime as part of DevOps team via Airflow orchestration.

FIDELITY INVESTMENTS, Bangalore, Karnataka

Software Engineer, 2016 – 2017

- Integrated 7 data source expanding the reach of the data management team.
- Increased quality gate coverage from 171 of 807 services using state of the art libraries.

EDUCATION

ARIZONA STATE UNIVERSITY, Tempe, Arizona - **M.S., Computer Science, 2017 - 2019**

NATIONAL INSTITUTE OF TECHNOLOGY, KARNATAKA, India - **B.S., Computer Science, 2012 - 2016**

PUBLICATIONS

[*A Novel Machine Learning Approach for Bug Prediction*](#), 2016

[*Comparison of OpenCL and CUDA frameworks on Heterogeneous Systems*](#), 2015

SKILLS

Languages: Python, Java, Type Script, JavaScript, C++, C

AI/ML: Lang Chain, Lang Graph, NLP, LLM, scikit-learn, Milvus, Vector Databases, RAG

Data bases: PostgreSQL, Cosmos DB, MongoDB, MSSQL, Elastic Search, Oracle

Cloud & Infra: Kubernetes, Docker, AWS, Azure, CI/CD, Kafka, Hadoop, Hashi Corp Vault, Akeyless

Frameworks: Fast API, React, Node.js, Express, Graph QL

Observability: Prometheus, Grafana, Splunk, Dynatrace, Open Telemetry

Architecture: Microservices, Distributed Systems, Event-Driven Pipeline

AWARDS

National Rated **Chess** Player - Ranked in the top **0.001%** at National level entrance examination. **Problem solver.**